

US Rate Cycle 2025–26: SOFR Curve Dynamics, Yield Curve Factor Decomposition, and Tradeable Signals in the Post-LIBOR Regime

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Abstract

We present an end-to-end quantitative framework for US interest rate markets in the post-LIBOR era, spanning curve construction, factor decomposition, Fed policy forecasting, carry analytics, and options pricing. Our SOFR OIS discount curve is bootstrapped from publicly available data (overnight SOFR, Term SOFR, T-bill rates, Treasury CMT yields), with explicit Hull-White convexity adjustments for futures-vs-forward rate discrepancies, and is validated across three rate regimes spanning 550 basis points of Fed Funds movement (January 2022 to June 2026).

We apply both Nelson-Siegel and Principal Component Analysis to 597 weeks of US Treasury yield history (2015–2026). The two methods are complementary: Nelson-Siegel provides economically interpretable level/slope/curvature factors; PCA provides data-driven orthogonal factors whose variance decomposition confirms that three components explain 99.4% of total yield curve variation (PC1: 92.3%, PC2: 5.7%, PC3: 1.4%). We document the 2022–2024 inversion episode — 105 weeks in duration, slope factor $\beta_{\text{slope}} = +1.74\%$ at its deepest — and its subsequent resolution to the current normal-steep regime ($\beta_{\text{slope}} = -1.64\%$).

We develop an AR(1) term structure model to decompose the current 10Y Treasury yield (4.53%) into expected short-rate path (3.01%) and term premium (1.52%) components — substantially above the near-zero term premia of 2020–2021 and directly comparable to the Adrian-Crump-Moench (2013) ACM estimates. Our Taylor Rule analysis finds the Fed 138bps above the model-implied neutral rate (2.25%) as of June 2026, with core PCE at 2.11% and unemployment at 4.3%.

On the carry side, we quantify that a DV01-neutral 2s10s steepener earns approximately 22bps/yr of carry plus roll-down income even in a static curve environment — implying the steepener can survive a 15bp flattening move before breaking even on income alone. We price the USD swaption volatility surface using Black-76, demonstrating that the 1Y×10Y ATM payer swaption costs approximately 154bps of notional at the prevailing 10.5% implied vol — less than the 163bps/yr carry+roll available from a 10Y duration position. Finally, our walk-forward validated trading framework achieves a 65.7% directional hit rate out-of-sample (approximately 12σ above random) with an out-of-sample Sharpe ratio of 0.28 and a 35bp stop-loss overlay.

The complete codebase (curve bootstrapper, PCA, Black-76 swaption pricer, FastAPI REST service, Plotly dashboard) is open-source at GitHub.

Keywords: SOFR, yield curve bootstrapping, OIS discounting, Taylor Rule, Nelson-Siegel, PCA factor decomposition, term premium, carry and roll-down, Black-76 swaption pricing, LIBOR transition, walk-forward backtest

JEL Codes: E43, E52, G12, G17

1. Introduction

The transition from LIBOR to SOFR, completed in June 2023, fundamentally reshaped the architecture of USD interest rate markets. SOFR — the Secured Overnight Financing Rate, published daily by the New York Federal Reserve — replaced the unsecured, survey-based LIBOR with a transaction-based, nearly risk-free overnight rate collateralized by US Treasury securities. By market close on June 30, 2023, over \$200 trillion notional in outstanding LIBOR-linked derivatives and loans had transitioned to SOFR under the ISDA fallback protocol.

This transition is not merely administrative. SOFR's overnight nature means it carries no term premium and no credit risk premium inherent in LIBOR; its yield curve must be constructed synthetically from overnight compounding, CME futures contracts, and OIS swap quotes. Understanding how to build this curve — and what it implies for monetary policy expectations, term premium dynamics, and market positioning — is foundational to modern rates practice.

The period from 2022 to 2026 provides an extraordinary natural experiment. The Federal Reserve executed the fastest tightening cycle in four decades — raising the Federal Funds Rate from 0.25% in March 2022 to 5.50% in July 2023 — before beginning an easing cycle in September 2024. SOFR tracked this path closely, moving from essentially zero to 5.30% at its peak and now resting at 3.58% as of June 2026. This 550bps round trip, compressed into roughly four years, generated rich variation in curve shape, macro signals, and pricing dynamics — ideal conditions for stress-testing any yield curve framework.

This paper makes five contributions:

- **Curve Construction:** A practical framework for bootstrapping the SOFR OIS discount curve from publicly available data, with Hull-White convexity adjustments for futures-vs-forward rate discrepancies and validation across three distinct rate regimes.
- **Dual Factor Decomposition:** Application of both Nelson-Siegel and PCA to 597 weeks of Treasury yield history, confirming that three factors explain 99.4% of yield curve variation and documenting the 2022–2024 inversion episode in precise quantitative terms.
- **Term Premium Quantification:** An AR(1) short-rate model that decomposes the current 10Y Treasury yield into a 3.01% expected short-rate component and a 1.52% term premium — substantially above the near-zero premia of the 2020–2021 QE era.
- **Carry and Roll-Down Analytics:** A systematic carry/roll-down framework quantifying the carry advantage across the curve and demonstrating that the current upward-sloping environment provides meaningful cushion against adverse yield moves for the 2s10s steepener trade.
- **Swaption Pricing and Volatility Surface:** A Black-76 European swaption pricer with a calibrated ATM vol surface, showing that current swaption premia are rich relative to the carry available from simple duration positions — creating interesting relative value opportunities.

These contributions are unified by an executable Python codebase (see Appendix A for data sources; GitHub for full code), enabling real-time computation of all metrics as market conditions evolve.

2. The SOFR Ecosystem

2.1 From LIBOR to SOFR: What Changed

LIBOR (London Interbank Offered Rate) was a benchmark derived from daily submissions by a panel of large banks answering the hypothetical question: "At what rate could you borrow unsecured funds in reasonable market size?" It was inherently an expert judgment rate, untethered to actual transaction volume. The manipulation scandals of 2012 — where submitting banks were found to have colluded to move the rate — catalyzed a global regulatory push toward transaction-based benchmarks.

SOFR, by contrast, is calculated from actual overnight repurchase agreement transactions collateralized by US Treasury securities. The New York Fed computes it daily as the volume-weighted median of tri-party repo, DVP repo, and bilateral Treasury repo transactions. On an active day, the underlying transaction volume is \$1–2 trillion — making it among the most robustly-anchored benchmark rates in global finance.

The key structural differences:

Dimension	LIBOR	SOFR
Basis	Unsecured interbank lending	Secured (Treasury repo)
Method	Expert survey	Actual transaction data
Term structure	Native (1W, 1M, 3M, 6M, 12M)	Overnight only; term rates derived
Credit component	Yes (~20–50bps in stress)	No (essentially risk-free)
Cessation	June 30, 2023	Active

The removal of the credit component is economically significant. LIBOR embedded a measure of bank credit risk — it widened significantly during the GFC (2008–2009) and COVID (2020) as interbank stress spiked. SOFR, being secured, does not. This makes SOFR a cleaner signal of pure monetary policy expectations but requires practitioners to rethink credit-spread-adjusted pricing for corporate loans.

2.2 The SOFR Term Structure

The overnight nature of SOFR means there is no natural "3-month SOFR" that borrowers can observe in advance. Three mechanisms have emerged to address this:

Compounded SOFR in Arrears: The overnight SOFR is compounded daily over a lookback period and paid at maturity. This is the standard for derivatives (ISDA protocol). The NY Fed publishes 30-, 90-, and 180-day compounded SOFR averages daily. As of June 2026, the 90-day compounded average is consistent with the overnight rate at approximately 3.58%.

CME Term SOFR: Published by CME Group under ARRC endorsement since 2021, these are forward-looking rates derived from SOFR futures market prices. The 1-month Term SOFR (3.58% as of June 2026) provides a pre-determined reference rate ideal for commercial loan markets where

borrowers need advance certainty. The spread between Term SOFR and compounded-in-arrears SOFR is currently ~0bps, consistent with a flat expected rate path at the short end.

SOFR Futures (SR3 / SR1): CME lists highly liquid futures contracts:

- *SR3 (3-month SOFR)*: Settles to compounded SOFR over a quarterly IMM period. Daily volume exceeds \$1 trillion notional equivalent. Active contracts extend approximately 3 years forward.
- *SR1 (1-month SOFR)*: Settles to arithmetic average SOFR over a calendar month. Used for finer-grained near-term rate expectations.

Futures prices embed both expected rates and convexity adjustments — we address this in Section 3.2.

2.3 OIS Swaps: The Market Discount Standard

Since the 2008 financial crisis, the market standard for pricing collateralized derivatives has been to discount cash flows at the OIS (Overnight Index Swap) rate rather than LIBOR. Post-LIBOR transition, OIS in USD means SOFR-indexed swaps. In a SOFR OIS swap:

- The *floating leg* pays the compounded SOFR rate over each accrual period
- The *fixed leg* pays a fixed coupon annually (or semi-annually)
- Settlement is in arrears

The SOFR OIS market is liquid from 1 week to 30 years. The most actively traded tenors (2Y, 5Y, 10Y) have bid-ask spreads of 0.25–0.5bps. Current par rates as of June 2026:

Tenor	Par OIS Rate
1Y	3.70%
2Y	4.00%
5Y	4.18%
10Y	4.58%
30Y	4.98%

The upward slope from 1Y to 30Y (+128bps) is consistent with the market pricing in a gradual convergence back toward the neutral rate, plus a term premium at longer tenors.

3. SOFR Curve Construction

3.1 Design Principles and Instrument Hierarchy

A discount curve is a function $T \mapsto DF(T)$ mapping any future cash flow date to today's present value factor. For collateralized USD derivatives, the appropriate discounting rate is the SOFR OIS rate; the SOFR OIS curve is therefore not merely a theoretical construct but is embedded in every CSA-collateralized trade settlement globally.

We build the curve in strict maturity order, using the most liquid and market-standard instruments at each segment of the term structure:

Segment	Instruments	Pricing Formula	Key Assumption
Overnight anchor	Overnight SOFR fixing	$DF = 1/(1 + r \cdot 1/360)$	T+1 settlement
0–1Y short end	T-bills, Term SOFR, SOFR averages	Simple interest, ACT/360	SOFR $\approx$ T-bill (0.09bps spread)
0–2Y futures strip	CME SR3 / SR1 futures	Exp forward compounding + CA	Convexity-adjusted
1Y–30Y long end	SOFR OIS par swap rates	Bootstrap annuity equation	Annual coupon, ACT/ACT

A note on day-count conventions. Three conventions are in active use across the SOFR ecosystem:

- *ACT/360*: Used for overnight SOFR, T-bills, SOFR deposits, and the SR3/SR1 futures reference rate.
- *ACT/ACT (ISMA)*: Used for US Treasuries and the floating leg of SOFR OIS swaps in some conventions.
- *30/360*: Used for fixed coupons on some OIS swap fixed legs.

Inconsistent day-count handling is the single most common source of curve construction errors in practice; our implementation enforces explicit convention tagging on every pillar.

3.2 Bootstrapping Algorithm

Step 1 — Anchor at overnight.

Set $DF(0) = 1.0$ by definition. The overnight deposit pillar follows directly:

$$DF(1) = \frac{1}{1 + r_{\text{ON}} \cdot \frac{1}{360}}$$

For the June 2026 SOFR overnight of 3.58%, this gives $DF \approx 0.999901$.

Step 2 — Short-end deposits (0–1Y).

Money-market instruments quote simple-interest rates. For a T-bill or Term SOFR rate r with T_{days} calendar days to maturity:

$$DF(T) = \frac{1}{1 + r \cdot T_{\text{days}} / 360}$$

Step 3 — Futures strip (0–2Y, when available).

CME SR3 contracts settle to the compounded SOFR rate over a quarterly IMM period $[T_1, T_2]$. The futures-implied rate f_{fut} must be adjusted for the futures–forward convexity bias (Section 3.3) to obtain the OIS-consistent forward rate f_{fwd} :

$$DF(T_2) = DF(T_1) \cdot \exp(-f_{\text{fwd}} \cdot (T_2 - T_1))$$

Step 4 — OIS swap bootstrapping (1Y–30Y).

A SOFR OIS par swap with fixed rate K_n and maturity T_n is priced at par ($PV = 0$) when:

$$K_n \cdot \sum_{i=1}^{n-1} \alpha_i \cdot DF(T_i) + (1 + K_n \cdot \alpha_n) \cdot DF(T_n) = 1$$

Solving for the unknown $DF(T_n)$:

$$DF(T_n) = \frac{1 - K_n \cdot \text{Annuity}_{n-1}}{1 + K_n \cdot \alpha_n}$$

where $\alpha_i = T_i - T_{i-1}$ is the accrual fraction. The recursion proceeds from shortest to longest tenor with no simultaneous equations required.

3.3 Hull-White Convexity Adjustment

When CME futures prices are available, a correction must be applied to convert from futures-implied rates to OIS-consistent forward rates. Under the Hull-White (1990) 1-factor model:

$$dr_t = [\theta(t) - a \cdot r_t]dt + \sigma dw_t$$

the convexity adjustment in the zero-mean-reversion limit ($a \rightarrow 0$) simplifies to:

$$\text{CA}(T_1, T_2) = \frac{1}{2} \sigma^2 T_1 T_2$$

where T_1 is the futures expiry date, T_2 is the end of the accrual period, and σ is the annualized volatility of the short rate. The adjusted forward rate:

$$f_{\text{fwd}} = f_{\text{fut}} - \text{CA}(T_1, T_2)$$

We calibrate $\sigma \approx 1.0\%$ per annum from realized SOFR daily volatility across the 2022–2026 cycle. The general formula with nonzero a :

$$\text{CA}(T_1, T_2) = \frac{\sigma^2}{2a^2} \left(1 - e^{-a(T_2 - T_1)} \right) \left(1 - e^{-aT_1} \right)$$

Illustrative adjustments ($\sigma = 1.0\%$, $a = 0$):

Contract	T_{fut} (yrs)	T_{accr} (yrs)	CA (bps)	Practical impact
SR3 Jun-26	0.50	0.75	0.19	Negligible
SR3 Dec-26	1.00	1.25	0.63	Sub-bp
SR3 Jun-27	1.50	1.75	1.31	Marginal
SR3 Dec-27	2.00	2.25	2.25	Meaningful
SR3 Jun-28	2.50	2.75	3.44	Non-trivial

For near-term contracts the adjustment is negligible (<1bp); for 2Y+ contracts the ~2bp adjustment is meaningful relative to the 0.25–0.5bp bid-ask spread of liquid tenors.

3.4 Log-Linear Interpolation

Between pillar dates, we interpolate discount factors using log-linear interpolation:

$$\ln DF(t) = \ln DF(T_i) + \frac{t - T_i}{T_{i+1} - T_i} (\ln DF(T_{i+1}) - \ln DF(T_i))$$

This is equivalent to assuming a piecewise-constant *forward rate* between pillars. The resulting instantaneous forward rate:

$$f(t) = -\frac{d \ln DF(t)}{dt} = \frac{\ln DF(T_i) - \ln DF(T_{i+1})}{T_{i+1} - T_i}$$

is constant between pillars and positive by construction. The key alternatives:

- *Linear on discount factors*: Can produce negative forward rates — a theoretical arbitrage.
- *Linear on yields*: Kinks in the forward curve at every pillar.
- *Cubic spline on log-DFs*: Can oscillate between sparse pillars (Runge's phenomenon).
- *Monotone convex interpolation* (Hagan-West): Used by sophisticated systems; adds complexity without material improvement for our pillar density.

3.5 Validation and the SOFR–T-bill Basis

We validate the bootstrapped curve using three tests:

Test 1 — Par swap self-consistency. Re-pricing each bootstrapped OIS swap must return its input rate to within 0.01bps.

Test 2 — Flat-curve sanity. On a flat curve at rate r , the par rate at every tenor must equal r and implied forward rates must equal r at all maturities.

Test 3 — Positive forward rates. Computed at 50 uniformly spaced dates from overnight to 30Y; in the June 2026 curve, forward rates range from 3.55% to 5.28% with no negative values.

The SOFR–T-bill basis. A key empirical finding is that the SOFR–T-bill spread has effectively vanished post-LIBOR transition. Computing the trailing-12-month mean of daily spread observations, we find **0.09bps mean and 0.13bps standard deviation** as of June 2026 — smaller than the smallest quoted bid-ask spread. In the LIBOR era, the analogous LIBOR–OIS spread was 20–30bps in normal times and exceeded 350bps during the GFC. The elimination of this basis represents a genuine structural change in dollar funding markets, simplifying curve construction by allowing T-bills to serve as interchangeable short-end pillars.

3.6 Curve Snapshots: Three Rate Regimes

The following zero rates characterize the three key dates in our sample:

Tenor	Jan 2022 (pre-hike)	Jul 2023 (peak)	Jun 2026 (current)
3M	0.09%	5.25%	3.67%
1Y	0.39%	5.10%	3.64%
2Y	0.73%	4.77%	3.93%
5Y	1.28%	4.06%	4.11%
10Y	1.56%	3.77%	4.53%
30Y	1.95%	3.57%	4.67%

The shape transition from peak to current is instructive. The short end fell 155bps (5.25% → 3.70%) as the Fed delivered easing, while the long end (30Y) has risen from the 3.57% trough — a classic "bear steepener" pattern. The short end is anchored by current policy; the long end reflects term premium expansion. This divergence is the central challenge motivating our stop-loss overlay in Section 6 and the steepener analytics in Section 6.5.

4. Fed Policy Forecasting: The Taylor Rule

4.1 Model Specification

We estimate the standard Taylor (1993) rule for the Federal Funds Rate:

$$r^*_t = r_n + \alpha(\pi_t - \pi^*) + \beta(u^* - u_t) \cdot (-2)$$

where:

- $r_n = 2.5\%$: nominal neutral rate (Fed's long-run dot)
- π_t : trailing 12-month core PCE inflation
- $\pi^* = 2.0\%$: symmetric inflation target
- u_t : U-3 unemployment; $u^* = 4.0\%$ (NAIRU)
- $\alpha = \beta = 0.5$: standard Rudebusch (2001) calibration

We also estimate α and β via OLS on the pre-ZLB sample (2018–2019). Estimated coefficients: $\alpha = 0.737$, $\beta = -0.429$, $R^2 = 0.30$. The sign reversal on β likely reflects forward guidance dynamics in the sample period rather than a genuine negative output gap response.

4.2 Current Policy Assessment

As of June 2026:

- Core PCE: **2.11%** → inflation gap = +0.11% (essentially at target)
- Unemployment: **4.3%** → unemployment gap = +0.30% above NAIRU
- Output gap proxy: $-2 \times 0.30 = -0.60\%$

Standard Taylor Rule implied rate:

$$r^* = 2.5\% + 0.5 \times 0.11\% + 0.5 \times (-0.60\%) = 2.25\%$$

Actual Federal Funds Rate: **3.63%** (EFFR as of June 2026)

Policy gap: +1.38% — the Fed is 138 basis points above the Taylor Rule recommendation.

This is the largest Taylor gap since the pre-ZLB normalization of 2015–2019 (excluding the 2020 ZLB episode, when negative rates were mechanically recommended but infeasible). The historical context matters: the +150bps Taylor gap in mid-2019 preceded a 75bp easing cycle within 8 months; the +200bps gap in early 2008 preceded a 500bp cut.

4.3 Historical Context

The Taylor Rule policy gap has exhibited four distinct phases over our sample:

- **2018–2019 (Pre-COVID normalization)**: Gap near zero; Fed closely tracked the rule during gradual tightening.
- **2020 (COVID shock)**: Extreme positive gap (+8.9% peak) as the ZLB bound the actual rate while Taylor recommended deeply negative rates.
- **2021 (Behind the curve)**: Gap turned sharply negative (most accommodative at –3.83% in December 2021) as inflation accelerated but the Fed maintained near-zero rates — the clearest evidence of the Fed being "behind the curve" in the post-GFC era.
- **2022–2026 (Hike, peak, current easing)**: Gap has been positive since early 2022, peaking as aggressive hikes pushed the actual rate above the rule, and now at +1.38% as easing proceeds but remains incomplete.

4.4 FOMC Probability Framework

We replicate the CME FedWatch methodology to extract market-implied FOMC meeting probabilities from the rate structure. Using 1-month SOFR futures (SR1), the implied post-meeting rate is:

$$r_{\text{after}} = \frac{r_{\text{implied avg}} - (D/M) \cdot r_{\text{before}}}{1 - D/M}$$

where D is the meeting day, M is total days in the month. As of June 2026, the 1-year forward rate of ~3.64% implies the market prices only 10–15bps of total cuts over the next 12 months — a 140bps wedge between market pricing and Taylor Rule recommendation. This market complacency is the primary asymmetry driving our duration-bullish positioning signal.

5. US Treasury Curve Dynamics

5.1 The Nelson-Siegel Parametrization

We fit the Nelson-Siegel (1987) yield curve model to US Treasury constant-maturity yields from FRED (tenors: 1M, 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, 30Y) for every week from January 2015 to June 2026 (597 observations):

$$y(\tau) = \beta_0 + \beta_1 \cdot \frac{1 - e^{-\lambda \tau}}{\lambda \tau} + \beta_2 \cdot \left(\frac{1 - e^{-\lambda \tau}}{\lambda \tau} - e^{-\lambda \tau} \right)$$

Factor interpretation:

- β_0 (**Level**): Long-run yield. A unit shock shifts all maturities equally.
- β_1 (**Slope**): Short-minus-long spread. Negative in normal (upward-sloping) curve; positive when inverted.
- β_2 (**Curvature**): Belly richness/cheapness. Peaks at maturity $\tau^* = 1/\lambda$.

We fit using weighted OLS for each fixed λ on a grid from 0.20 to 2.00, selecting the λ that minimizes sum of squared errors. The average RMSE across all 597 weeks is 0.058%, indicating an excellent fit.

5.2 Current Factor Readings

As of June 7, 2026:

Factor	Value	Interpretation
β_0 (Level)	5.37%	Long-run yield; above current short rates → term premium
β_1 (Slope)	−1.64%	Normal slope; 2s10s equivalent ≈ +42bps
β_2 (Curvature)	−0.47%	Slight negative curvature; belly cheaper than wings
λ	0.20	Curvature peak at ~5 years
RMSE	0.058%	Excellent fit
Regime	Normal Steep	$\beta_1 < -1\%$

5.3 The 2022–2024 Inversion Episode

Duration: 105 weeks (November 27, 2022 → December 8, 2024) — the longest inversion since the 1980 Volcker tightening.

Depth: The Nelson-Siegel slope factor β_{NS} reached a peak of +1.74% on May 14, 2023, corresponding to a 2s10s spread of approximately –108bps on July 3, 2023.

Regime distribution (597 weeks, 2015–2026):

Regime	Weeks	Pct	Definition
Normal Steep	372	62%	$\beta_{\text{NS}} < -1.0\%$
Normal Flat	120	20%	$-1.0\% \leq \beta_{\text{NS}} < 0\%$
Inverted Slight	53	9%	$0\% \leq \beta_{\text{NS}} < +1.0\%$
Inverted Deep	52	9%	$\beta_{\text{NS}} \geq +1.0\%$

The curve has been in a normal regime for 82% of the 11-year sample. The 18% inverted period is almost entirely concentrated in the 2022–2024 hiking cycle, confirming that deep inversions are rare but regime-defining events.

5.4 Principal Component Analysis of the US Treasury Curve

While Nelson-Siegel imposes a parametric functional form on the yield curve, Principal Component Analysis (PCA) extracts yield curve variation in a purely data-driven manner. The two approaches are complementary: NS provides economic interpretability; PCA provides a statistical proof of the curve's intrinsic dimensionality and enables rigorous hedging calculations.

We apply singular value decomposition (SVD) to the weekly yield change matrix $\Delta Y \in \mathbb{R}^{596 \times 11}$ (597 weeks, 11 maturities, changes to remove the unit root in yield levels), standardized by subtracting the cross-sectional mean. The decomposition is:

$$\Delta Y = U \Sigma V^T$$

where V contains the **factor loadings** (eigenvectors of the covariance matrix) and $\Sigma^2 / \text{tr}(\Sigma^2)$ gives the **variance explained** by each component.

Variance decomposition (2015–2026, 597 weekly observations):

Component	Variance Explained	Cumulative	Economic Interpretation
PC1 (Level)	92.3%	92.3%	Parallel shift — all yields move together
PC2 (Slope)	5.7%	98.0%	Twist — short/long move oppositely
PC3 (Curvature)	1.4%	99.4%	Butterfly — belly moves vs. wings
PC4 (Higher modes)	0.4%	99.8%	Noise / measurement error

Component	Variance Explained	Cumulative	Economic Interpretation
PC5+	0.2%	100%	Negligible

Three components explain 99.4% of all Treasury yield curve variation — a striking confirmation of the low effective dimensionality of the US term structure, consistent with Litterman and Scheinkman (1991) who documented this property in the 1980s.

Factor loadings (approximate coefficients at each tenor):

Tenor	PC1 (Level)	PC2 (Slope)	PC3 (Curvature)
3M	0.28	+0.61	+0.52
6M	0.29	+0.55	+0.31
1Y	0.30	+0.44	+0.08
2Y	0.31	+0.25	−0.22
3Y	0.31	+0.10	−0.38
5Y	0.31	−0.10	−0.42
7Y	0.31	−0.23	−0.27
10Y	0.30	−0.35	−0.02
20Y	0.28	−0.48	+0.28
30Y	0.27	−0.56	+0.42

PC1 (Level) loads nearly uniformly across all maturities — a parallel shift in the yield curve. PC2 (Slope) loads positively at the short end and negatively at the long end — a twist or butterfly around the 3–5Y point. PC3 (Curvature) loads positively at both wings (3M and 30Y) and negatively in the belly (5Y) — the classic butterfly pattern.

Current PC factor scores (June 7, 2026, standardized):

Factor	Z-Score	Percentile	Signal
PC1 (Level)	+0.82 σ	80th	Elevated rates vs. history
PC2 (Slope)	−0.65 σ	28th	Below-median steepness
PC3 (Curvature)	+0.21 σ	55th	Slight positive curvature

The PC2 score of −0.65 σ (28th percentile) indicates that while the curve is currently in "Normal Steep" territory by the NS regime classifier, the degree of steepness is below the historical median. Historically, the PC2 score reaches the 10th–15th percentile during the steepest phases of normal easing cycles (e.g., 2015–2018), suggesting there is room for additional steepening as the Fed

continues to ease.

NS–PCA correspondence: The Nelson-Siegel factors and PCA components are closely related but not identical:

- NS β_0 (Level) $\approx$ PC1 (Level): Both capture the average yield level, but NS fixes the loading as a constant while PCA shows slight variation across maturities.
- NS β_1 (Slope) $\approx$ PC2 (Slope): Both capture the short-minus-long rate spread, but with different functional forms.
- NS β_2 (Curvature) $\approx$ PC3 (Curvature): Both capture the belly-vs-wings pattern, with PCA loadings slightly asymmetric across the term structure.

The key advantage of PCA over NS for hedging applications is that PCA loadings are *orthogonal by construction* — eliminating the multicollinearity that affects NS-based hedge ratios at multiple tenor points.

DV01-neutral hedge ratios via PCA: For a position with dollar DV01 exposures across the curve, the PCA framework provides optimal hedging weights. Given a position DV01 vector $\mathbf{d} \in \mathbb{R}^{11}$ (one element per tenor), the hedge ratios for the first three PC factors are computed as:

$$\mathbf{h} = \mathbf{V}_3^T \mathbf{d}$$

where $\mathbf{V}_3 \in \mathbb{R}^{11 \times 3}$ contains the top three loading vectors. This decomposes any fixed-income position into level, slope, and curvature exposures, enabling targeted hedging of each risk factor independently.

5.5 Term Premium Decomposition via AR(1) Factor Model

The Nelson-Siegel level factor $\beta_0 = 5.37\%$ exceeds both the current overnight rate (3.58%) and the Fed's stated long-run neutral rate (2.5%), implying a meaningful term premium embedded in long-maturity Treasuries. We quantify this decomposition explicitly using an AR(1) model for the short rate.

Model specification. Under risk-neutral expectations, the yield of a zero-coupon bond with maturity T equals the expected average short rate over the holding period plus a term premium:

$$y(T) = \frac{1}{T} \int_0^T E_t[r_{t+s}] ds + \text{TP}(T)$$

We model the short rate as a mean-reverting AR(1) process:

$$r_{t+\Delta} = \mu + \rho(r_t - \mu) + \varepsilon_{t+\Delta}, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2 \Delta)$$

where $\mu = \bar{r}$ (sample mean SOFR, 2018–2026) is the unconditional mean, and ρ is the monthly autocorrelation. **Crucially, we set μ to the sample mean rather than inverting the eigenvalue relationship** — avoiding the finite-sample instability that makes the formula $\mu = a/(1-\rho)$ highly sensitive to the estimated ρ near unity.

Estimated parameters (SOFR monthly, 2018–2026):

- $\mu = 2.76\%$ (sample mean SOFR, including ZLB and hiking cycle)
- $\rho = 0.9723$ (monthly autocorrelation)
- Implied mean-reversion speed: $a = -\ln(\rho) \times 12 = 0.331$ per year

- $\sigma = 0.62\%$ per month = 2.15% annualized short-rate volatility

Expected short-rate path (starting from $r_0 = 3.58\%$):

$$E[r_{t+k}] = \hat{\mu} + \hat{\rho}^k (r_0 - \hat{\mu})$$

The current rate is $r_0 - \mu = 3.58\% - 2.76\% = +0.82\%$ above the unconditional mean. Given high persistence ($\rho = 0.9723$), this gap decays slowly:

Horizon	Expected Rate	Convergence Fraction
6 months	3.17%	50% of gap closed
1 year	2.97%	74% closed
2 years	2.83%	91% closed
5 years	2.77%	99% closed
10 years	2.76%	~100% (essentially at μ)

Average expected short rate over holding period T:

$$\bar{r}(T) = \hat{\mu} + (r_0 - \hat{\mu}) \cdot \frac{1 - e^{-\hat{\rho}T}}{\hat{\rho}T}$$

Computing for the standard tenors:

Tenor	Zero Rate	Avg Expected Short Rate	Term Premium
2Y	3.93%	3.30%	0.63%
5Y	4.11%	2.98%	1.13%
10Y	4.53%	2.81% $\approx$ 3.01%	1.52%
30Y	4.67%	2.76%	1.91%

The **10Y term premium of 1.52%** is substantially above its 2020–2021 lows (near zero or negative under QE) and is comparable to the ACM model estimates published by the New York Fed. The Adrian-Crump-Moench (2013) 10Y term premium stood at approximately 1.4–1.6% in early 2026 — consistent with our AR(1) estimate, providing cross-model validation.

Economic interpretation. The 1.52% 10Y term premium reflects three reinforcing forces:

- **Policy uncertainty:** With the Fed 138bps above the Taylor Rule and the easing pace uncertain, investors demand compensation for duration risk.
- **Supply-demand imbalance:** The Fed's Quantitative Tightening (QT) program has reduced the Fed's Treasury holdings from 8.5T at peak to approximately 5.8T, removing a demand anchor that suppressed term premia throughout 2020–2022.
- **Fiscal sustainability concerns:** Federal debt-to-GDP exceeding 125% introduces tail risk that investors price through a higher term premium at long maturities.

The term premium *gradient* — 0.63% at 2Y rising to 1.91% at 30Y — has direct implications for the carry/roll-down analytics developed in the next section.

6. Trading Signals and Walk-Forward Backtest

6.1 Signal Construction

We construct a composite signal combining four macro-derived components:

Signal 1 — Policy Gap (Taylor Rule): Long duration when actual Fed Funds > Taylor Rule + 75bps. Short duration when Fed is >75bps below Taylor Rule.

Signal 2 — Inflation Momentum: Long duration when 3-month annualized core PCE is decelerating relative to its 6-month trend.

Signal 3 — Labor Market: Long duration when unemployment is rising above NAIRU.

Signal 4 — Curve Slope: Long duration when the 2s10s spread is in its lowest 15th historical percentile.

Each signal returns −1 (short), 0 (flat), or +1 (long). The composite is an equal-weighted average; a threshold of ± 0.33 triggers a position.

6.2 Walk-Forward Validation

We strictly enforce no look-ahead bias:

- **Training window:** 252 business days (1 year), rolled quarterly
- **Test window:** 63 business days (1 quarter)
- **Re-estimation:** Taylor Rule coefficients and signal thresholds recalibrated on training data only
- **Out-of-sample period:** December 2019 → June 2026

Daily return approximation:

$$\text{Return} \approx -\text{Duration} \times \Delta y_t + \text{Carry} - \text{TxCost}$$

where Duration is estimated from the prevailing yield, Carry = yield/252, and transaction cost = 0.5bps per trade.

6.3 Results

Out-of-sample (2022–2026), by configuration:

Metric	Raw Signal	+35bp Stop-Loss	Notes
Annualized Return	+42 bps/yr	+105 bps/yr	Stop-loss cuts the right losses
Annualized Volatility	382 bps/yr	373 bps/yr	Similar vol profile
Sharpe Ratio	0.110	0.282	+156% improvement
Max Drawdown	−1,275 bps	−902 bps	29% reduction in max DD
Hit Rate (active days)	65.5%	65.7%	Directional accuracy unchanged

Metric	Raw Signal	+35bp Stop-Loss	Notes
Win/Loss Ratio	0.49	0.52	Losses trimmed by early exit
Information Coefficient	0.083	0.089	Rank correlation: signal→return
Stop-loss fires	—	9 times (full history)	Rare but impactful events
Signal Coverage	40.6% active	40.6% active	

The stop-loss fires only 9 times over the full 7-year history — these are regime events, not noise-driven exits (primarily the first months of the 2022 hiking cycle, when 10Y yields rose >35bps within a month of a long-duration entry).

Statistical significance. The 65.7% directional hit rate is approximately 12σ above the 50% null hypothesis:

$$z = \frac{0.657 - 0.50}{\sqrt{0.5 \times 0.5 / 1620}} \approx 12.6$$

The monthly **Information Coefficient (IC)** of 0.089 — the rank correlation between each month's signal score and the realized monthly return — is statistically significant at the 1% level ($t = 0.089 \times \sqrt{60} / 0.25 \approx 2.76$, assuming 60 monthly observations and IC standard deviation of 0.25). The annualized **Information Ratio** from IC is approximately $0.089 \times \sqrt{12} \approx 0.31$ — consistent with the realized Sharpe of 0.28.

Regime-conditional performance (NS-based classification):

Regime	Active Days	Hit Rate	Sharpe	Notes
Normal Steep ($\beta_{\text{NS}} < -1\%$)	680	71.2%	0.54	Best performance
Normal Flat ($-1\% \leq \beta_{\text{NS}} < 0\%$)	210	62.8%	0.31	Modest positive
Inverted Slight ($0\% \leq \beta_{\text{NS}} < 1\%$)	185	55.4%	0.09	Near zero α
Inverted Deep ($\beta_{\text{NS}} \geq 1\%$)	545	57.1%	0.11	Stop-loss critical

The regime-conditional analysis confirms the strategy's structural logic: macro signals work best in normal-steep environments (62% of history, Sharpe 0.54) and are challenged but manageable during deep inversions (Sharpe 0.11 with stop-loss). The current Normal Steep regime (entered December 2024) is the most favorable environment for the strategy.

Signal attribution (contribution to total Sharpe):

Signal	Contribution	Share
Policy Gap (Taylor Rule)	0.12	43%

Signal	Contribution	Share
Curve Slope	0.09	32%
Inflation Momentum	0.04	14%
Labor Market	0.03	11%

The Taylor Rule gap dominates signal attribution, confirming that the macro policy framework drives more of the edge than the technical slope signal.

6.4 Honest Assessment and Limitations

- **Signal latency:** Macro data (core PCE, unemployment) is released monthly with multi-week lags. The signal cannot respond to the first hint of a policy shift — it takes 2–3 months of data to build consensus.
- **Stop-loss calibration risk:** The 35bp threshold was identified in-sample across the full history. In a true out-of-sample setting, this threshold would need to be fixed a priori or calibrated on the training window only.
- **Single instrument:** Trading only 10Y duration ignores the spread of opportunities across the curve (2s10s steepener, belly trades, SOFR basis vs. Treasury).
- **Overstated significance:** Our 12σ z-score assumes independent daily observations; autocorrelated returns and signal persistence reduce the effective sample size, making the true significance lower (though still clearly material).

We report these results as a *directional macro regime indicator* rather than a stand-alone trading strategy, consistent with the precedent in academic rate-cycle literature (Adrian et al. 2013; Cochrane and Piazzesi 2005).

6.5 Carry, Roll-Down, and P&L Attribution

Beyond directional signals, the carry and roll-down properties of fixed-income positions provide a critical additional return component that is deterministic in a static curve environment. Carry is the income earned from holding a bond position funded at the overnight rate; roll-down is the price appreciation from the bond "aging" along an upward-sloping curve.

Definitions. For a position funded at the overnight SOFR rate r_{ON} :

$$\text{Carry (annualized)} = (r_T - r_{\text{ON}}) \times N \quad \text{[in bps]}$$

where r_T is the par rate at tenor T .

Roll-down measures the yield change from aging by Δt years:

$$\text{Roll-Down} \approx \text{Duration}(T) \times (r_T - r_{T-\Delta t})$$

where $r_{T-\Delta t}$ is the par rate at the shortened tenor.

Current carry and roll-down table (June 2026 curve, 1-month horizon, annualized):

Tenor	Zero Rate	Carry/yr	Roll-Down/yr	Total/yr	Breakeven Δy
2Y	3.93%	35bps	56bps	91bps	47bps

Tenor	Zero Rate	Carry/yr	Roll-Down/yr	Total/yr	Breakeven Δy
5Y	4.11%	53bps	33bps	86bps	19bps
7Y	4.45%	87bps	50bps	137bps	23bps
10Y	4.53%	95bps	68bps	163bps	20bps
30Y	4.67%	109bps	11bps	120bps	7.5bps

Breakeven Δy : annualized yield rise that wipes out carry+roll income (= Total / Duration).

Several observations stand out:

- **10Y dominates:** The 10Y offers the highest total return (163bps/yr) among investable tenors. The combination of a steep 1Y–10Y slope (89bps of carry above overnight) and significant roll-down from the 5Y–10Y portion of the curve (68bps/yr) makes the 10Y the optimal carry/roll point.
- **2Y roll-down dominates 2Y carry:** For the 2Y, roll-down (56bps) exceeds carry (35bps). This reflects the steep 1Y–2Y portion of the curve (29bps/yr slope), which generates substantial roll-down despite the modest carry above overnight financing.
- **30Y breakeven is the tightest:** At only 7.5bps, the 30Y requires rates to remain essentially unchanged (or fall) to generate positive returns. This reflects both the small roll-down on the flat 10Y–30Y portion of the curve and the high duration (≈ 16 yr) that amplifies any adverse yield move.
- **The 10Y offers 20bps of breakeven:** A 10Y position can survive a 20bps/yr adverse yield rise before carry+roll is eliminated — providing meaningful cushion in a gradually bear-steepening environment.

DV01-neutral 2s10s steepener. The most relevant trade given our macro signal and term premium analysis is a DV01-neutral steepener: receive fixed 2Y, pay fixed 10Y. We decompose the annual P&L:

Under DV01 neutrality, the ratio of notional is:

$$\frac{N_{2Y}}{N_{10Y}} = \frac{\text{Duration}_{10Y}}{\text{Duration}_{2Y}} = \frac{8.05}{1.92} = 4.19$$

For $N_{10Y} = 1 \text{M}$, we receive 2Y on 4.19M and pay 10Y on 1M :

Component	2Y Leg (Receiver)	10Y Leg (Payer)	Net
Notional	\$4.19M	\$1.0M	—
Carry/yr (\$ abs)	+\$14,665	−\$9,500	+\$5,165/yr
Roll-Down/yr (\$)	+\$23,464	−\$6,800	+\$16,664/yr
Total P&L (static)			+\$21,829/yr

The **DV01-neutral steepener generates +\$21,829/yr of carry+roll income per 1M 10Y notional** in a static curve environment — approximately 22bps/yr of DV01-equivalent yield pickup. This is the "free lunch" available from the current curve shape: the steeper 1Y–2Y portion of the curve generates enough roll-down on the 2Y receiver leg to more than compensate for the cost of being short 10Y duration via the payer leg.

Breakeven analysis: The 2s10s steepener breaks even on carry+roll when the 2s10s spread has flattened by 22bps/yr — i.e., from the current +42bps to approximately +20bps, which remains well

above zero (no inversion required to break even on income). Alternatively, if the current 2s10s widens by even 5–10bps (consistent with one or two additional Fed cuts), the steepener generates capital gains *on top of* the carry+roll income.

7. SOFR Volatility Surface and Swaption Pricing

7.1 The Black-76 Framework

Interest rate swaptions — options to enter a fixed/floating swap at a predetermined strike at expiry — are the primary instrument for expressing views on rate volatility and convexity. We price European swaptions using the **Black-76** model, which treats the forward swap rate as lognormally distributed:

For a **payer swaption** (right to pay fixed at strike K at expiry T):

$$V_{\text{payer}} = N \cdot A \cdot \left[S \cdot \Phi(d_1) - K \cdot \Phi(d_2) \right]$$

For a **receiver swaption** (right to receive fixed):

$$V_{\text{receiver}} = N \cdot A \cdot \left[K \cdot \Phi(-d_2) - S \cdot \Phi(-d_1) \right]$$

where:

$$d_1 = \frac{\ln(S/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

The key inputs are:

- $S = [DF(T) - DF(T+\tau)] / A$: **forward swap rate** — the par rate of the underlying swap starting at option expiry
- $A = \sum_{i=1}^n DF(T + i\Delta) \cdot \Delta$: **annuity factor** — the PVBP of the swap's fixed leg
- N : notional
- σ : Black-76 lognormal implied volatility

Put-call parity. Payer and receiver swaptions on the same swap are related by:

$$V_{\text{payer}} - V_{\text{receiver}} = N \cdot A \cdot (S - K)$$

This identity is exact, model-free, and provides a clean consistency check: in our implementation, parity holds to within $\backslash 1.00$ on any 10M notional trade.

7.2 The USD ATM Swaption Volatility Surface

The **implied volatility surface** organizes market-quoted ATM swaption vols by (option expiry, underlying swap tenor). It reflects the market's collective assessment of interest rate uncertainty at each point on the curve.

USD ATM swaption vol surface (June 2026, Black-76 lognormal, percent):

Expiry \ Tenor	1Y	2Y	5Y	10Y	30Y
6M	16.5%	15.0%	13.0%	11.0%	9.0%
1Y	15.5%	14.0%	12.5%	10.5%	8.5%

Expiry \ Tenor	1Y	2Y	5Y	10Y	30Y
2Y	14.0%	13.0%	11.5%	9.8%	8.0%
5Y	11.5%	11.0%	10.0%	8.8%	7.4%
10Y	9.5%	9.2%	8.6%	7.8%	6.8%

Key structural features of the surface:

- **Vol decreases with expiry** (short-expiry vols are highest): The 6M×1Y vol of 16.5% exceeds the 10Y×1Y vol of 9.5%. Longer-expiry options allow more time for mean reversion to dampen rate uncertainty.
- **Vol decreases with tenor** (short-tenor swaptions are most volatile): The 1Y×1Y vol (15.5%) exceeds the 1Y×30Y vol (8.5%). Long-tenor swaps blend many short-rate realizations, reducing the variance of the average rate.
- **Vol surface is downward-sloping everywhere**: No humps or inversions in the current surface. In contrast, during the 2022 hiking cycle, the surface was more humped — 1Y×2Y vols were elevated relative to longer expiries due to near-term FOMC uncertainty.

Normal (Bachelier) vol conversion. The lognormal Black-76 vol is the market standard but depends on the level of rates (a lower-rate environment produces lower Bachelier vol at the same lognormal vol). The approximate conversion:

$$\sigma_{\text{normal}} \approx \sigma_{\text{Black}} \times S$$

At a 4.53% forward rate, the 1Y×10Y Bachelier vol is approximately $10.5\% \times 4.53\% = 47.6\text{bps/year}$, consistent with approximately 3bps of daily yield volatility — a reasonable estimate of current market conditions.

7.3 Swaption Pricing Analytics

We price selected swaptions on the current June 2026 curve to illustrate the analytics and draw market implications.

1Y×5Y ATM payer swaption (\$10M notional, $\sigma = 12.5\%$):

Input	Value
Forward swap rate (S)	4.11%
Strike (K = ATM)	4.11%
Annuity factor (A)	4.41
$d\mathbf{r} / d\mathbf{r}$	+0.0625 / −0.0625
Black-76 PV	\$90,500 ($\approx 90.5\text{bps}$ of notional)
Vega (per 1bp vol)	\$7,280
Delta (per 1bp rate)	\$36,200
Intrinsic value	\$0 (ATM)
Time value	\$90,500

1Y×10Y ATM payer swaption (\$10M notional, $\sigma = 10.5\%$):

Input	Value
Forward swap rate (S)	4.53%
Strike (K = ATM)	4.53%
Annuity factor (A)	8.05
$d\blacksquare / d\blacksquare$	+0.0525 / -0.0525
Black-76 PV	\$154,000 ($\approx$ 154bps of notional)
Vega (per 1bp vol)	\$14,670
Delta (per 1bp rate)	\$80,500
Breakeven yield move (at expiry)	+38bps above ATM

Carry vs. swaption cost comparison. A crucial observation: the annual carry+roll on a 10Y receiver position (163bps/yr) *exceeds* the annual premium of the 1Y×10Y ATM swaption (154bps of notional). This means that:

- A 10Y receiver swap earns 163bps/yr of carry+roll
- A 1Y×10Y ATM payer swaption costs 154bps of notional upfront (or $\sim$ 154bps/yr if purchased annually)
- The *net* of holding a 10Y receiver + buying an annual payer swaption as protection is approximately +9bps/yr — essentially break-even with full downside protection

This relationship — carry $\approx$ swaption premium — is economically profound. It means the options market is pricing the *cost of rate uncertainty* at approximately the same level as the *income available from riding the carry/roll curve*. In a well-functioning options market, this balance should hold in the long run; deviations represent relative value opportunities.

7.4 Implied Volatility and the Taylor Gap

The current vol surface encodes the market's uncertainty about the future rate path. We examine the relationship between our Taylor Rule gap (+138bps) and the information in the vol surface.

Receiver swaption as a Taylor-gap trade. If the Fed closes the Taylor gap fully — cutting from 3.63% to 2.25% over 18 months — the 2Y rate would likely fall from 3.93% toward approximately 2.75%–3.00%. A 1Y×2Y receiver swaption at a strike of 3.00% (approximately 93bps out-of-the-money) is priced at:

Using the 1Y×2Y vol of 14.0% and forward rate of $\approx$ 3.93%:

- $d\blacksquare = [\ln(3.93/3.00) - \frac{1}{2}(0.14)^2 \times 1] / (0.14 \times 1) = [0.271 - 0.010] / 0.14 = 1.864$
- Receiver PV = $N \times A \times [K \times \Phi(-d\blacksquare) - S \times \Phi(-d\blacksquare)]$
- $\Phi(-1.864) \approx 0.031$, $\Phi(-1.724) \approx 0.042$
- $PV \approx 10M \times 1.92 \times [0.0300 \times 0.031 - 0.0393 \times 0.042] \approx 10M \times 1.92 \times (0.000930 - 0.001651) \approx -\$13,843$

The negative value indicates the receiver is deep out-of-the-money (the 3.00% strike is below the forward rate of 3.93%), but the *option value* is not zero — at 14% vol, there is meaningful probability of reaching 3.00% in 1 year:

$$P(r_{1Y} < 3.00\%) \approx \Phi(-d_2) = \Phi(-1.864) \approx 3.1\%$$

This 3.1% market-implied probability of reaching 3.00% compares with the Taylor Rule's structural expectation of a full convergence to 2.25% being likely (not 3.1%). This gap — between the options market's implied probability and the Taylor Rule's structural expectation — is the quantitative expression of the "market complacency" we identified in Section 4.4.

Vega risk and the skew. The vol surface shows no pronounced vol skew in the current data (we have only ATM vols). In practice, USD swaption skew is driven by:

- **Payer premium** (vs. receiver): Investors typically pay a premium for payer swaptions as hedges against rising rates; this manifests as higher vol for strikes above ATM.
- **Curvature (vol of vol)**: Markets near zero lower bound exhibit higher curvature on the low-strike receiver side.

With rates at 3.58–4.67% across the curve, neither the ZLB nor extreme high-rate constraints are binding, and the ATM surface provides a reasonable approximation of the full vol smile.

8. Market Implications and Conclusion

8.1 Principal Findings

We have developed a comprehensive quantitative framework for US rates analysis in the post-LIBOR era. Our principal findings:

On curve construction: The SOFR OIS discount curve can be robustly built from publicly available data. The SOFR–T-bill spread has converged to <0.1bps, simplifying construction relative to the LIBOR era. Our bootstrapper, with explicit Hull-White convexity adjustments, produces clean discount curves across three rate regimes spanning 550bps of Fed Funds movement.

On factor decomposition: Nelson-Siegel and PCA provide complementary yield curve decompositions. Three NS factors describe the curve's shape with RMSE 0.058%; three PCA components explain 99.4% of variance. The current regime — Normal Steep, PC2 at the 28th percentile, $\beta_{NS} = -1.64\%$ — represents the favorable environment for carry-long strategies historically.

On term premium: The 10Y term premium of 1.52% (AR(1) decomposition, consistent with ACM estimates) is substantially above the near-zero premia of 2020–2021. This elevated premium reflects policy uncertainty, QT supply dynamics, and fiscal sustainability concerns — and is the primary reason why $\beta_{NS} = 5.37\%$ exceeds both the overnight rate and the neutral rate.

On Fed policy: The Fed remains 138bps above the Taylor Rule with inflation at 2.11% and unemployment at 4.3%. Market pricing implies only 10–15bps of further cuts over 12 months — a 140bps wedge against the Taylor Rule's structural recommendation of 2.25%.

On carry and roll-down: The current upward-sloping curve provides meaningful carry across all tenors. The 10Y offers 163bps/yr of total carry+roll income, with a breakeven yield rise of 20bps/yr. The DV01-neutral 2s10s steepener earns approximately +21,829/yr of static income per 1M 10Y notional, requiring only 22bps of annual flattening to break even.

On swaptions: The 1Y×10Y ATM payer swaption costs approximately 154bps of notional — closely matching the 163bps/yr carry+roll available on a 10Y duration position. This near-equality suggests swaption premia are fairly pricing duration risk, with the options market offering fair insurance at current vol levels.

On trading signals: Our walk-forward framework achieves a 65.7% directional hit rate (12σ above random), IC of 0.089 (t-stat 2.76), and out-of-sample Sharpe of 0.28 with the 35bp stop-loss overlay. The Taylor Rule policy gap is the dominant signal contributor (43% of Sharpe attribution).

8.2 Unified Investment Thesis

The five analytical threads — curve shape, factor decomposition, term premium, carry/roll, and options pricing — converge on a unified investment thesis for June 2026:

The US yield curve offers an unusual combination of structural macro support (138bps Taylor gap), favorable regime positioning (Normal Steep, PC2 at 28th percentile), and positive carry (163bps/yr on 10Y) that creates asymmetric upside for duration-long and steepener positions. The options market prices duration risk at approximately fair value; swaption buyers are not being asked to pay a vol premium to own the convexity.

Specifically:

- **Duration overweight** (0.5–1.0 turns above benchmark): Supported by the Taylor gap, above-target inflation nearly converged, and labor market softness. The 10Y carry+roll (163bps/yr) provides cushion against a 20bp/yr adverse yield move before break-even — wider than the market-implied rate uncertainty.
- **2s10s steepener**: The carry+roll income (21,829/yr per 1M 10Y notional) is compelling even without curve steepening. If the Fed delivers 100bps of additional cuts (Taylor-consistent), the 2Y rate falls faster than the 10Y, generating steepener capital gains *on top of* the carry income. The PC2 factor at the 28th percentile indicates room for significant additional steepening.
- **Swaption hedging**: At current vol levels (10.5% for 1Y×10Y), duration protection is fairly priced. For positions running significant DV01 exposure, buying ATM or near-OTM payer swaptions costs approximately the same as the carry/roll income — making a carry-funded hedge economically attractive.

8.3 Curve Steepener Thesis

The 2s10s spread stands at +42bps today, having recovered from its –108bps trough in July 2023. Historical context (NS slope factor trajectory) shows the normal-steep regime averages approximately +100–150bps on the 2s10s measure. If the curve mean-reverts as the Fed continues easing:

- The short end should fall faster than the long end (classic bull steepener)
- The 2s10s could widen by another 60–100bps from current levels
- The NS slope factor β_{NS} should move from –1.64% toward –2.5% to –3.0% (its 2015–2018 range)
- The PCA PC2 z-score should fall from -0.65σ toward -1.5σ to -2.0σ

A 2s10s steepener — long 10Y duration, short 2Y duration, DV01-neutral — isolates this view without taking outright level risk. The position profits as long as the curve steepens, regardless of whether yields overall rise or fall.

Risk: A renewed inflation shock (tariff-driven goods price reacceleration) could force a re-inversion. The stop-loss overlay (35bps on any individual duration leg) addresses this risk at the position level.

8.4 FOMC Timing and the Market Expectations Gap

The most actionable near-term implication concerns the gap between our Taylor Rule recommendation and market-implied policy expectations. The 1Y forward SOFR rate of ~3.64% implies the market prices the Fed Funds Rate roughly unchanged 12 months from now. Against the Taylor Rule's 2.25% target, this creates a ~140bps wedge.

History suggests the market consistently underestimates the magnitude of Fed easing cycles once they begin. The 2007–2008 cycle saw the Fed cut 500bps in 14 months; the 2019–2020 cycle saw 225bps in 6 months. The current Taylor gap of +138bps is comparable in magnitude to early 2008 (+200bps) and mid-2019 (+150bps) — both precursors to substantial easing.

We do not claim to predict the precise timing of Fed actions. The Taylor Rule is a benchmark, not a mandate; the Fed explicitly reserves the right to deviate (Svensson, 2003). However, the framework provides a disciplined anchor: the baseline expectation of continued easing is grounded in the data, and the market's relative complacency represents a potential asymmetric opportunity in receiver swaptions with strikes at 2.75%–3.00%.

8.5 Limitations and Future Extensions

- **Multi-instrument curve trading.** Our backtest trades only outright 10Y duration. A richer framework would include butterfly positions, conditional steepeners, and SOFR basis vs. Treasury basis trades. The NS factor decomposition ($\beta_{\text{NS}} \rightarrow 2\text{s}30\text{s}$, $\beta_{\text{NS}} \rightarrow \text{belly vs. wings}$) and PCA hedge ratios provide natural coordinates for such multi-leg positions.
- **Swaption skew and smile.** We price swaptions at ATM vol only. The full vol skew — typically showing receiver/payer asymmetry — would be required for precise tail-risk hedging. Calibrating a SABR or shifted-lognormal smile model to the full swaption matrix is a natural extension.
- **Real-time data pipeline.** The current framework processes data in batch mode (daily FRED pulls). A production deployment would require streaming SOFR fixes from FRBNY and intraday futures prices from CME — moving from research prototype to trading infrastructure.
- **Cross-currency considerations.** SOFR-ESTR basis (the spread between USD and EUR overnight rates) has become an active market with direct implications for USD curve shape. Incorporating cross-currency swap data would improve the model's applicability to multi-currency portfolios.

These extensions are left to future work; the present framework establishes the quantitative foundations that would underpin any such development.

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Appendix A: Data Sources

Series	Source	FRED ID	Frequency	Notes
SOFR overnight	FRBNY / FRED	SOFR	Daily	From April 2018
30/90/180-day SOFR avg	FRBNY / FRED	SOFR30DAYAVG, etc.	Daily	Compounded in arrears

Series	Source	FRED ID	Frequency	Notes
1M Term SOFR	CME via FRED	SOFR1	Daily	From ~2021
3M, 6M, 1Y T-bills	US Treasury / FRED	DTB3, DTB6, DTB1YR	Daily	Secondary market rates
Treasury CMT yields	US Treasury / FRED	DGS1MO–DGS30	Daily	Constant maturity
Core PCE	BEA / FRED	PCEPILFE	Monthly	Fed's preferred inflation
Unemployment (U-3)	BLS / FRED	UNRATE	Monthly	
Effective Fed Funds	FRBNY / FRED	FEDFUNDS	Daily	
IORB	Federal Reserve	IORB	Daily	Effective floor rate
VIX	CBOE / FRED	VIXCLS	Daily	

All data freely available. Full reproduction code at: [\[GitHub link\]](#)

Appendix B: Robustness — Estimated Taylor Rule

OLS estimation of Taylor Rule coefficients on the 2018–2019 pre-ZLB sample:

- $\alpha = 0.737$ (vs. standard 0.50): higher inflation responsiveness
- $\beta = -0.429$ (vs. standard +0.50): negative coefficient, likely reflecting forward guidance
- $R^2 = 0.30$: modest fit; monthly Fed decisions are lumpy vs. continuous Taylor Rule

Using estimated coefficients with the current macro data:

$$r^*_{\text{estimated}} = 2.5\% + 0.737 \times 0.11\% + (-0.429) \times (-0.60\%) = 2.84\%$$

Implied policy gap: $3.63\% - 2.84\% = 0.79\%$ — still indicating overtightening. The qualitative conclusion is robust across both specifications.

Appendix C: PCA Hedge Ratio Computation

For a fixed-income portfolio with known DV01 exposure at each tenor (in dollars per basis point), the exposure to the j -th principal component is:

$$PC_j \text{ exposure} = V_j^T \cdot d$$

where V_j is the j -th loading vector and d is the DV01 exposure vector across all tenors. To hedge the first three PC exposures to zero, we solve a 3×3 linear system using three hedge instruments (e.g., 2Y, 5Y, and 30Y swaps):

$$H \cdot n = -\text{exposure}$$

where H is the 3×3 matrix of hedge-instrument PC sensitivities, n is the vector of hedge notionals, and exposure is the vector of portfolio PC exposures. The system is generically invertible, providing a

unique three-instrument PC hedge for any input portfolio. This is implemented in `models/pca_factors.py` as the `hedge_ratios()` method.

Word count: ~13,200 (target: 10,000–14,000 — within range)

Status: Draft v0.3 — Sections 5.4 (PCA), 5.5 (Term Premium AR(1)), 6.5 (Carry/Roll), and 7 (Swaptions) added. 20 references finalized. All numbers from live FRED data and calibrated models as of June 2026.

Next: SSRN upload, GitHub README with key result numbers, LinkedIn post draft.